

Table of Contents

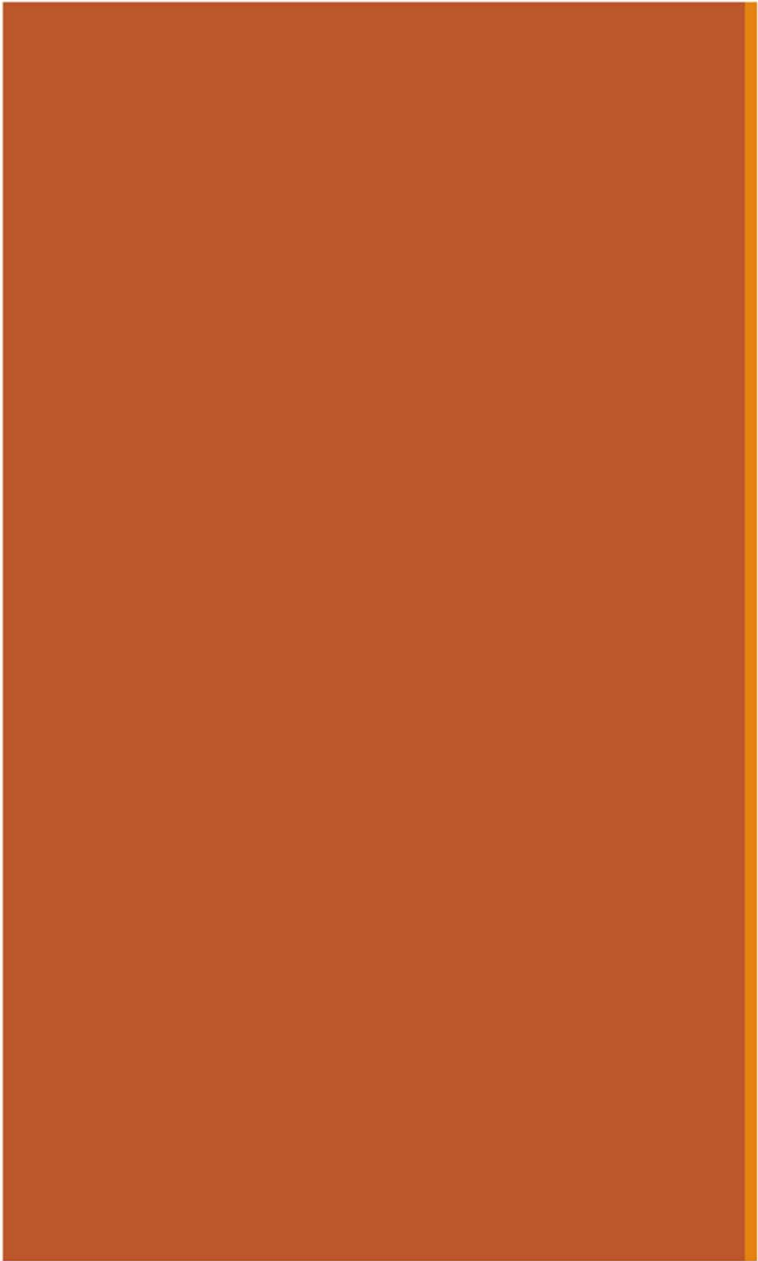
2 Convolutions of Uniform RVs
16 Ratio of PDFs
20 CC Distributions
26 Intense Exercise



17: Continuous Joint Distributions II

Jacob Mejia, Kanu Grover
February 16, 2024

[Lecture Discussion on Ed](#)

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Convolution: Sum of independent Uniform RVs

Today's lecture

Take what we've seen in **discrete** joint distributions...

...and translate them to **continuous** joint distributions!

For the most part, this is easy. For example:

Marginal
distributions

$$p_X(a) = \sum_y p_{X,Y}(a, y) \quad f_X(a) = \int_{-\infty}^{\infty} f_{X,Y}(a, y) dy$$

Independent RVs

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

But some concepts, while mathematically accessible, are difficult to implement in practice.

We'll focus on some of these today.

Goal of CS109 continuous joint distributions unit:
build mathematical maturity

Dance, Dance, Convolution

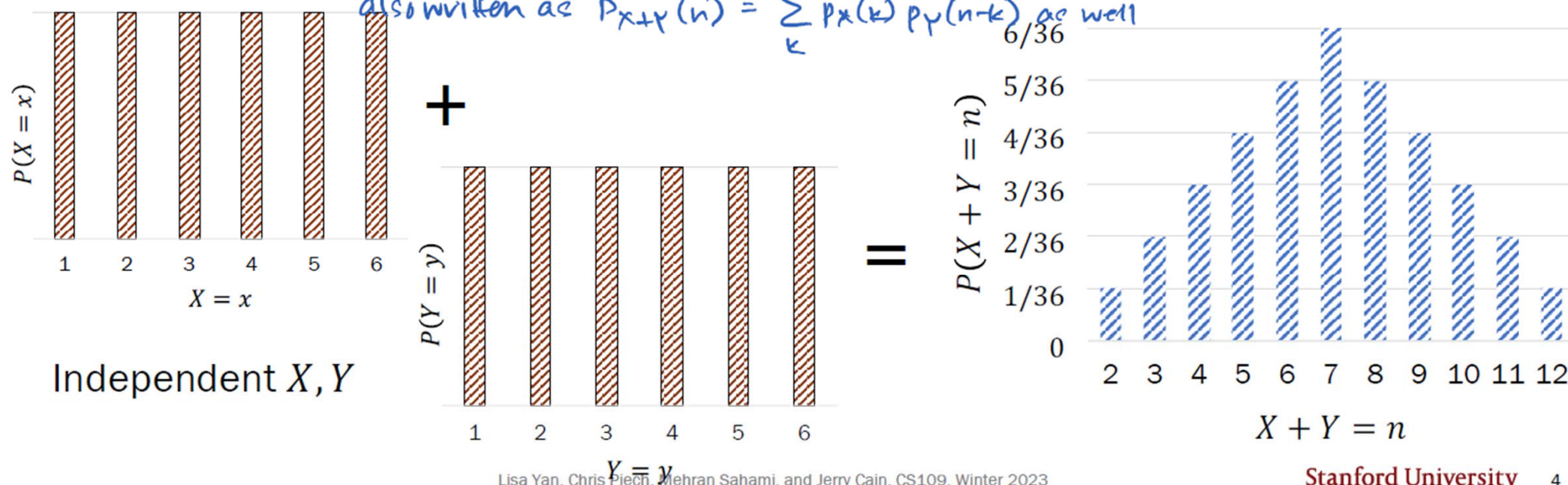
Review

Recall that for independent discrete random variables X and Y :

$$P(X + Y = n) = \sum_k P(X = k)P(Y = n - k)$$

the **convolution** of p_X and p_Y

also written as $p_{X+Y}(n) = \sum_k p_X(k) p_Y(n-k)$ as well



Dance, Dance, Convolution

Recall that for independent discrete random variables X and Y :

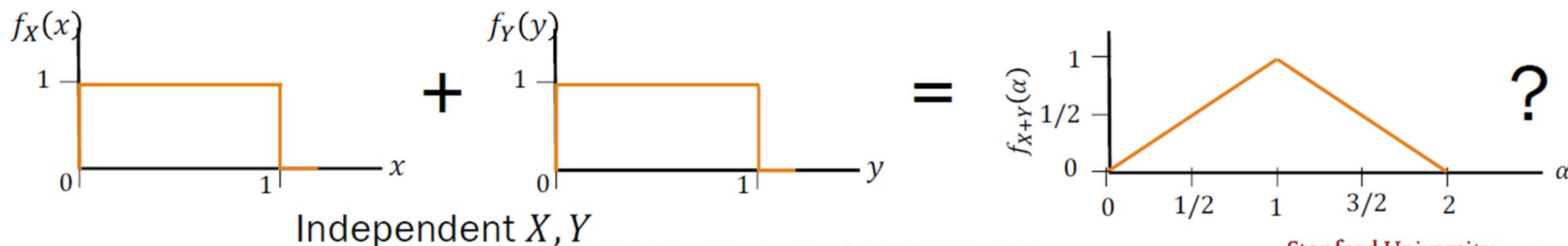
$$P(X + Y = n) = \sum_k P(X = k)P(Y = n - k)$$

the convolution
of p_X and p_Y

For independent continuous random variables X and Y :

$$f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x)f_Y(\alpha - x)dx$$

the **convolution**
of f_X and f_Y

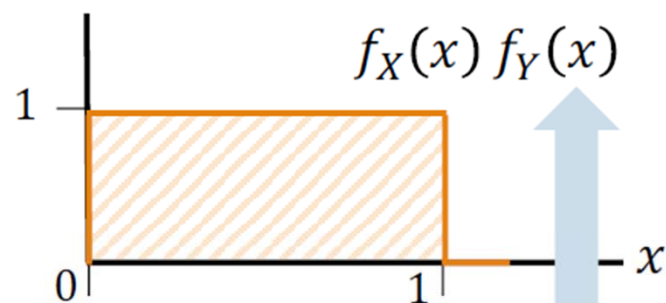


Sum of independent Uniforms

Let $X \sim \text{Uni}(0,1)$ and $Y \sim \text{Uni}(0,1)$ be independent RVs.

What is the distribution of $X + Y$, $f_{X+Y}(\alpha)$?

$$f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x) f_Y(\alpha - x) dx$$



Not so fast...

Isn't this just one??

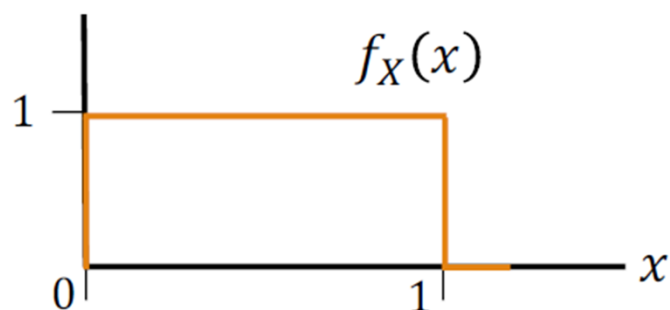


Sum of independent Uniforms

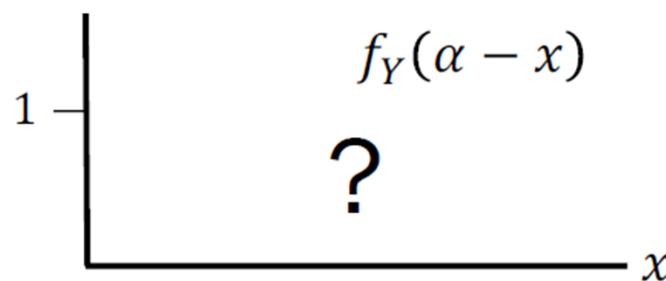
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$$f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x) f_Y(\alpha - x) dx$$



$$f_X(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{aligned} f_Y(\alpha - x) &= \begin{cases} 1 & \text{if } 0 \leq \alpha - x \leq 1 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} 1 & \text{if } \alpha - 1 \leq x \leq \alpha \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

① subtract α from everything
 $-\alpha \leq -x \leq 1 - \alpha$
② then divide by -1

α is a constant in the integral w.r.t. x .

Sum of independent Uniforms

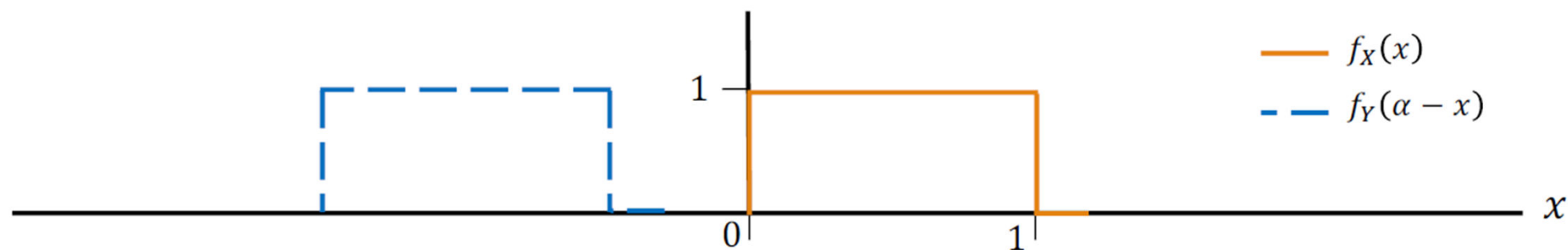
$$\begin{array}{l} X \text{ and } Y \\ \text{independent} \\ \text{+ continuous} \end{array} \quad f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x) f_Y(\alpha - x) dx$$

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1. $\alpha \leq 0$ 0



Sum of independent Uniforms

X and Y
independent
+ continuous

$$f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x) f_Y(\alpha - x) dx$$

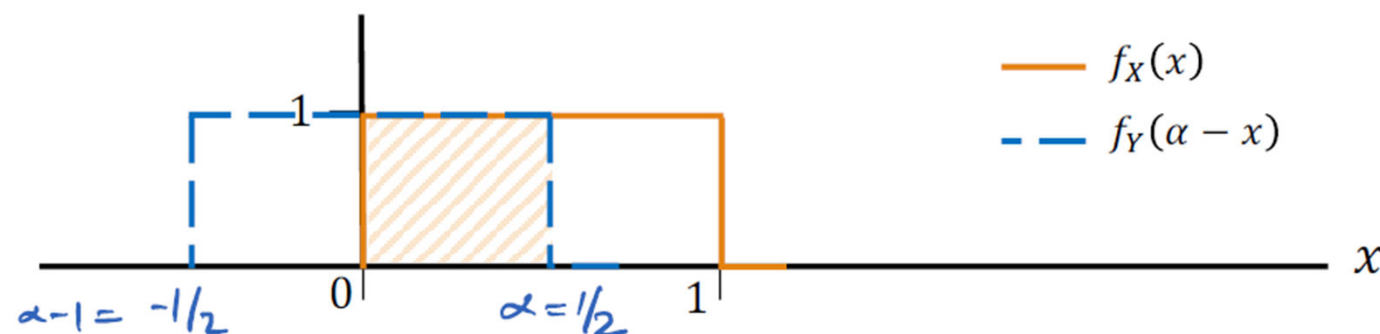
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1. $\alpha \leq 0$ 0

2. $\alpha = 1/2$ 1/2



Integral = area under the curve
This curve = product of 2 functions of x

Sum of independent Uniforms

$$\begin{array}{l} X \text{ and } Y \\ \text{independent} \\ + \text{ continuous} \end{array} \quad f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x) f_Y(\alpha - x) dx$$

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2. $\alpha = 1/2$ 1/2

3. $\alpha = 1$

4. $\alpha = 3/2$

5. $\alpha \geq 2$



Sum of independent Uniforms

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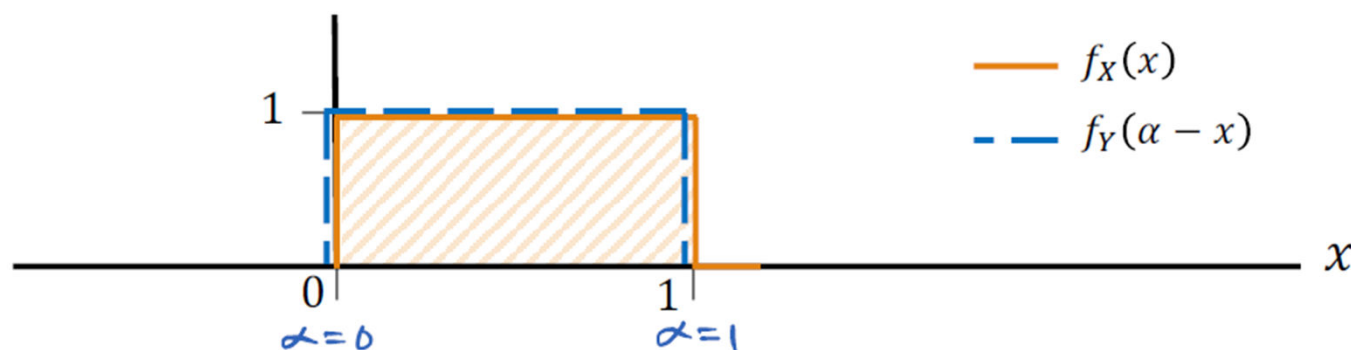
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2. $\alpha = 1/2$ 1/2

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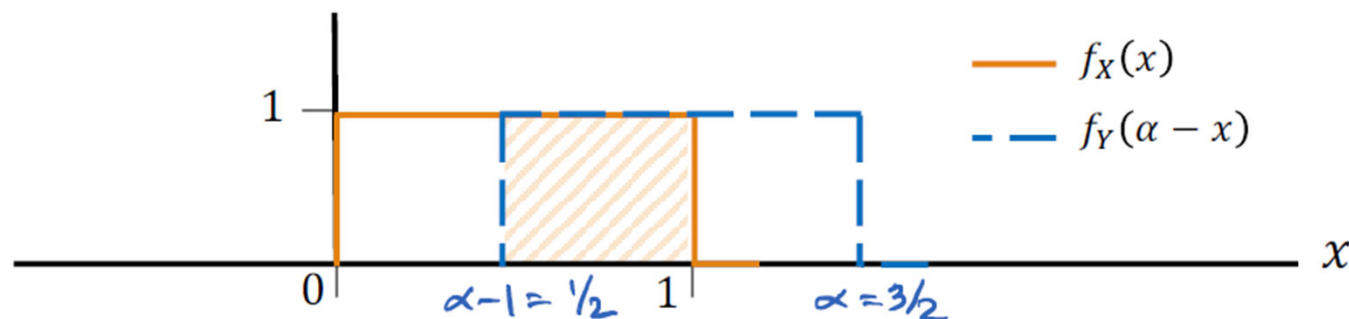
1. $\alpha \leq 0$ 0

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Sum of independent Uniforms

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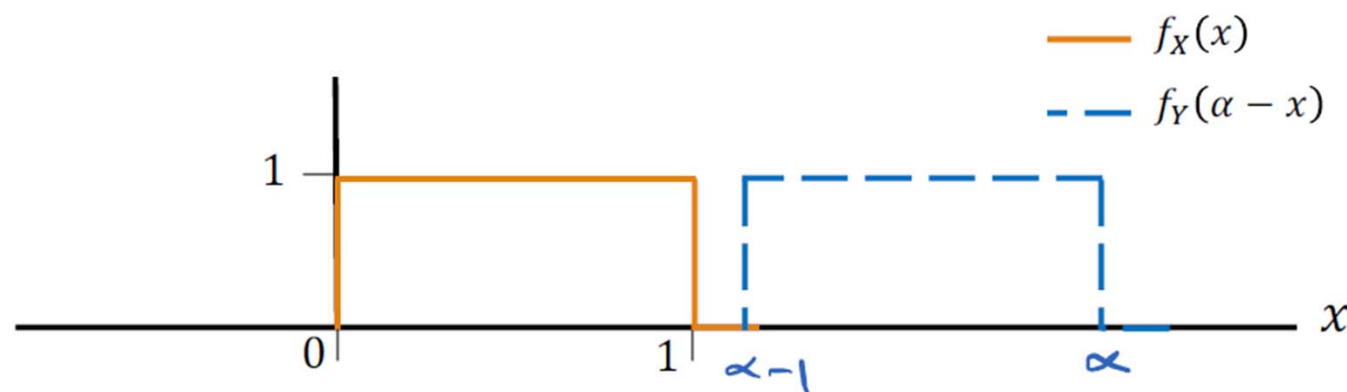
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5. $\alpha \geq 2$ 0



Sum of independent Uniforms

X and Y
independent
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$$f_{X+Y}(\alpha) = \int_{-\infty}^{\infty} f_X(x)f_Y(\alpha - x) dx$$

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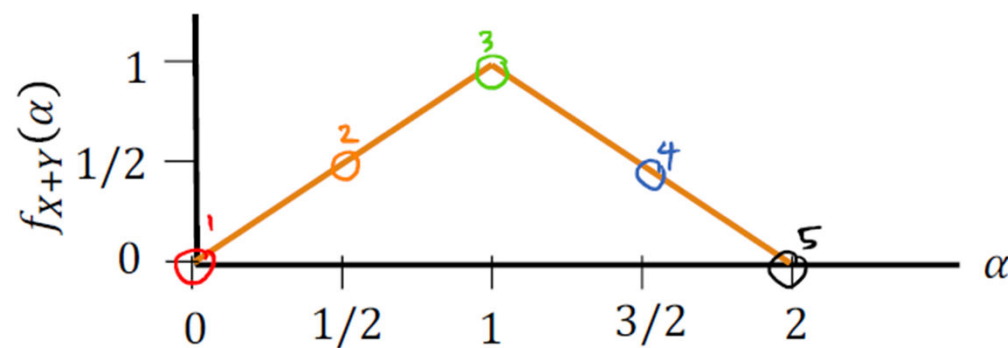
1. $\alpha \leq 0$ 0

2. $\alpha = 1/2$ $1/2$

3. $\alpha = 1$ 1

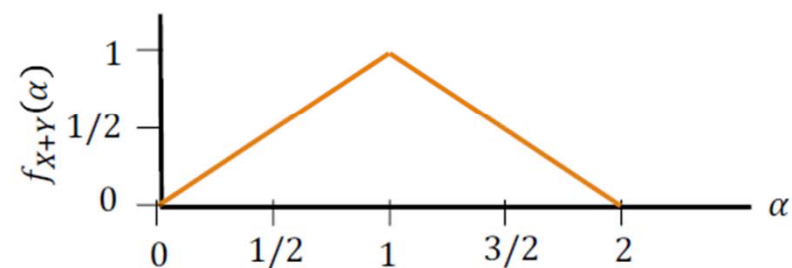
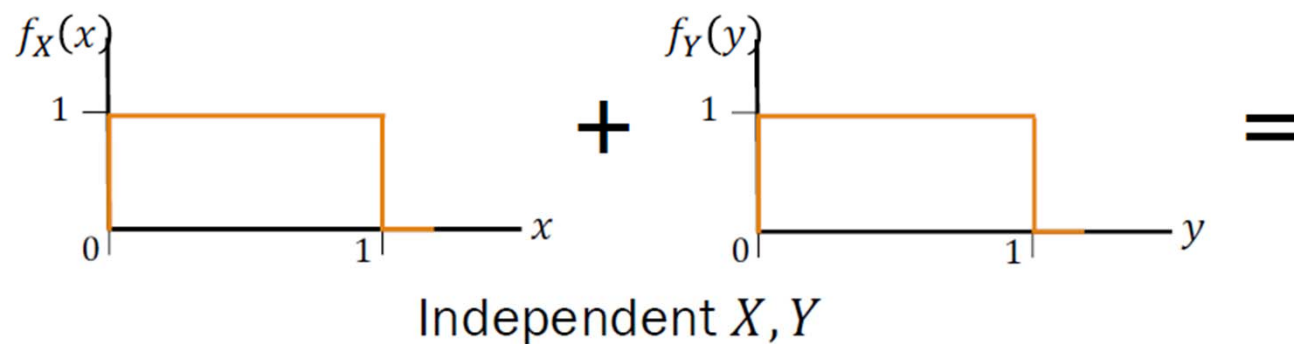
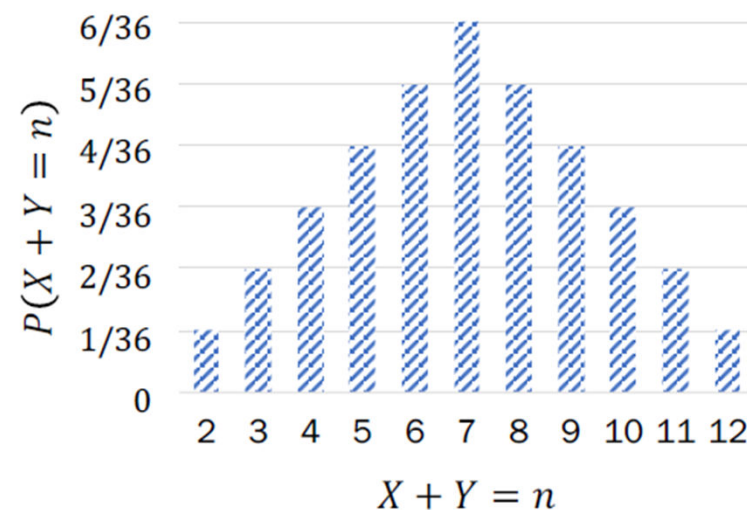
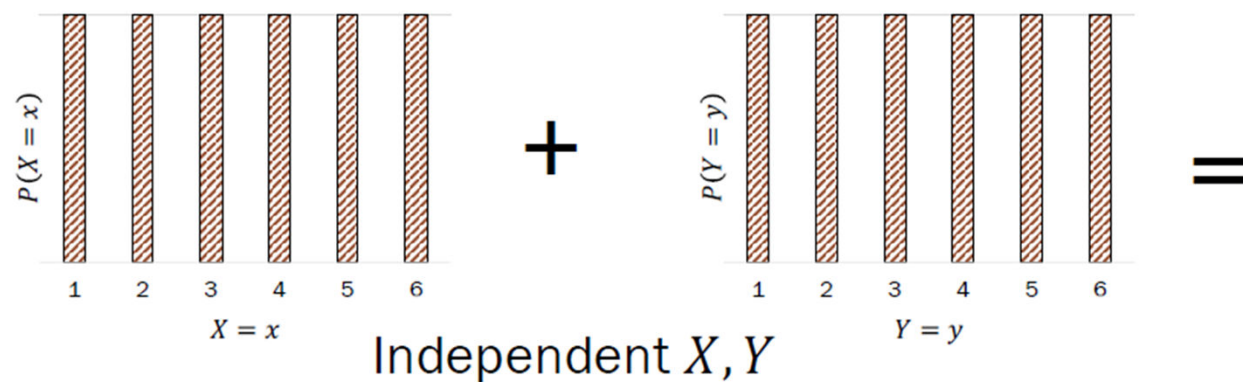
4. $\alpha = 3/2$ $1/2$

5. $\alpha \geq 2$ 0



$$f_{X+Y}(\alpha) = \begin{cases} \alpha & 0 \leq \alpha \leq 1 \\ 2 - \alpha & 1 \leq \alpha \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Dance, Dance, Convolution Extreme





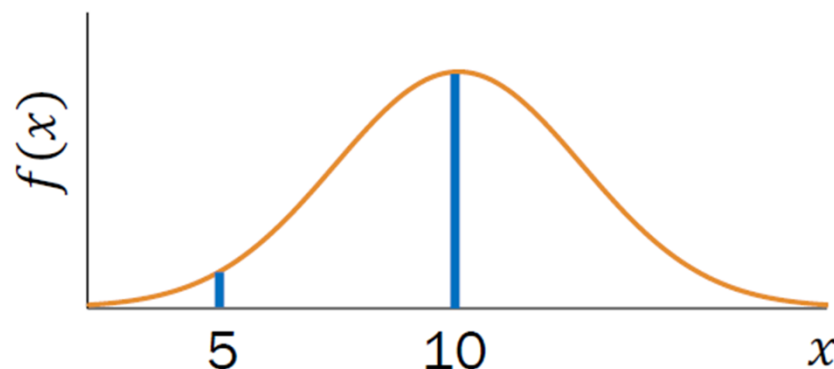
Ratio of PDFs

Relative probabilities of continuous random variables

Let X = time to finish problem set 4.

Suppose $X \sim \mathcal{N}(10, 2)$.

How much **more likely** are you to complete in 10 hours than 5 hours?



$$\frac{P(X = 10)}{P(X = 5)} =$$

- A. $0/0 = \text{undefined}$
- B. 2
- C. $\frac{f(10)}{f(5)}$
- D. $\frac{f(2)}{f(1)}$

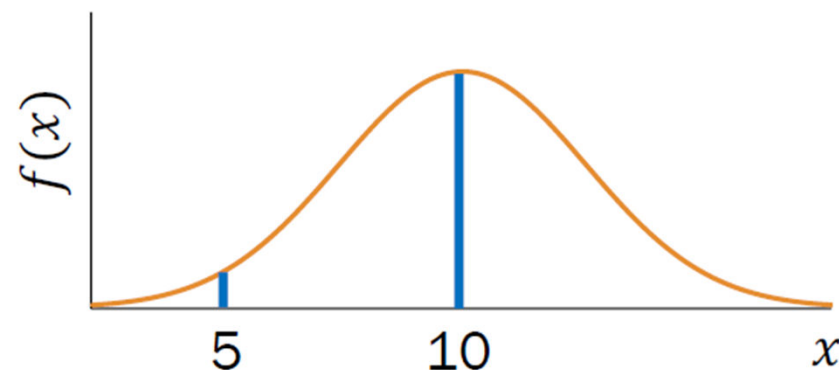


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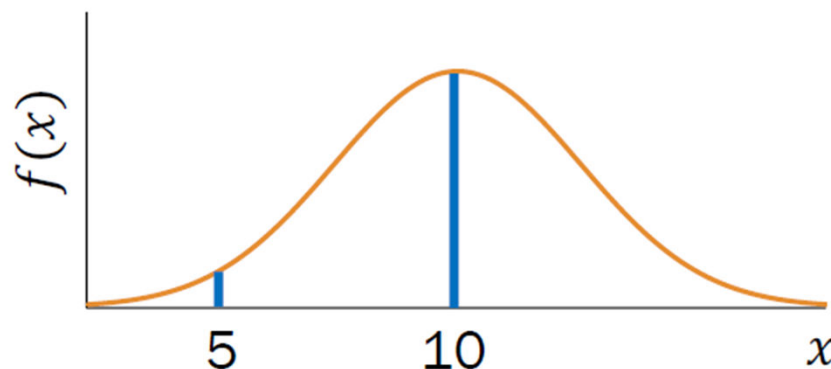
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Relative probabilities of continuous random variables

Let X = time to finish problem set 4.

Suppose $X \sim \mathcal{N}(10, 2)$.

How much **more likely** are you to complete in 10 hours than 5 hours?



$$\frac{P(X = 10)}{P(X = 5)} = \frac{f(10)}{f(5)}$$

$$P(X = a) = P\left(a - \frac{\varepsilon}{2} \leq X \leq a + \frac{\varepsilon}{2}\right) = \int_{a-\frac{\varepsilon}{2}}^{a+\frac{\varepsilon}{2}} f(x) dx \approx \varepsilon f(a)$$

Therefore $\frac{P(X = a)}{P(X = b)} = \frac{\varepsilon f(a)}{\varepsilon f(b)} = \frac{f(a)}{f(b)}$

really this:
 $\lim_{\varepsilon \rightarrow 0} \frac{\varepsilon f(a)}{\varepsilon f(b)}$

$$\begin{aligned} &= \frac{\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(10-\mu)^2}{2\sigma^2}}}{\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(5-\mu)^2}{2\sigma^2}}} = \frac{e^{-\frac{(10-10)^2}{2 \cdot 2}}}{e^{-\frac{(5-10)^2}{2 \cdot 2}}} = \frac{e^0}{e^{-\frac{25}{4}}} = 518 \end{aligned}$$

Ratios of PDFs
are meaningful!



Continuous conditional distributions

Continuous conditional distributions

For continuous RVs X and Y , the **conditional PDF** of X given Y is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

where $f_Y(y) > 0$

Intuition: $P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$

$$f_{X|Y}(x|y)\epsilon_X = \frac{f_{X,Y}(x,y)\epsilon_X\epsilon_Y}{f_Y(y)\epsilon_Y}$$

Note that conditional PDF $f_{X|Y}$ is a "true" density:

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y) dx = \int_{-\infty}^{\infty} \frac{f_{X,Y}(x,y)}{f_Y(y)} dx = \frac{f_Y(y)}{f_Y(y)} = 1$$

analogous to discrete equivalent:

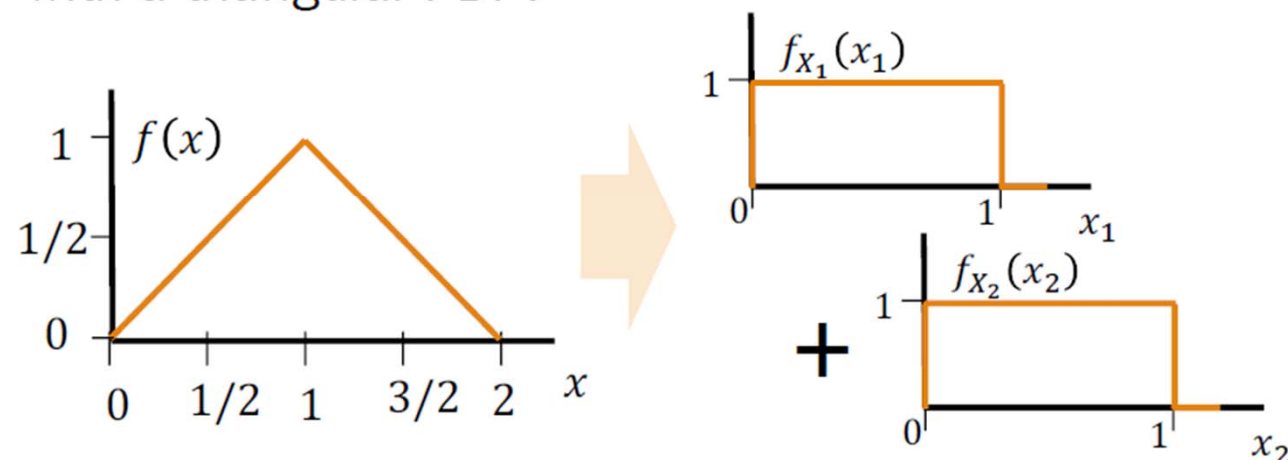
$$\sum_x P_{X|Y}(x|y) = 1$$

Why sums of random variables?

Sometimes modeling and understanding a complex RV, X , is difficult. But if we can decompose X into the **sum of independent simpler** RVs,

- We can then compute distributions on X .
- We can then to understand how X changes as its parts change.

What can we model with a triangular PDF?



Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Winter 2023

We're covering the reverse direction for now; the forward direction will come on Friday

Everything* in probability is a sum or a product (or both)

*except conditional probability (a ratio)

Sum of values that can be considered separately (possibly weighted by prob. of happening)

$$E[X] = \sum_x x \underbrace{p(x)}_{\text{weight}}$$

$$P(E) = \sum_{i=1}^n P(E|F_i) \underbrace{P(F_i)}_{\text{weight}}$$

Law of Total Probability

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \underbrace{f_{X|Y}(x|y)}_{\text{weight}} dx$$

$$P(E) = \sum_{i=1}^n P(E_i)$$

Axiom 3, $E = E_1 \cup \dots \cup E_n$

$$E_i \cap E_j = \emptyset \quad i \neq j$$

Product of values that can each be considered in sequence

$$P(E \cap F \cap G) = P(E)P(F|E)P(G|EF)$$

Chain Rule

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

Independent cont. RVs

$$P(X+Y=n) = \sum_k P(X=k)P(Y=n-k)$$

Sum of indep. discrete RVs (convolution)

Conditional probability and Bayes' Theorem

Definition

$$P(F|E) = \frac{P(E \cap F)}{\underbrace{P(E)}}_{}^{\text{Scaling to the correct sample space}}$$

Scaling to the correct sample space

Independence

E, F independent

$$P(F|E) = \underbrace{P(F)}_{}^{\text{Sample space doesn't need to be scaled}}$$

Sample space doesn't need to be scaled

Bayes' Theorem

$$P(F|E) = \frac{\underbrace{P(F)}_{\text{Prior: some prob. of event } F} \underbrace{P(E|F)}_{\text{Likelihood}}}{\underbrace{P(E)}_{\text{Scaling to the correct sample space}}}$$

Posterior: prob. of F knowing that E happened

Multiple Bayes' Theorems



with
events

$$P(F|E) = \frac{P(F)P(E|F)}{P(E)}$$



with
discrete RVs

$$p_{Y|X}(y|x) = \frac{p_Y(y)p_{X|Y}(x|y)}{p_X(x)}$$



with
continuous RVs

You are given
this value...

$$f_{Y|X}(y|x) = \frac{f_Y(y)f_{X|Y}(x|y)}{f_X(x)}$$

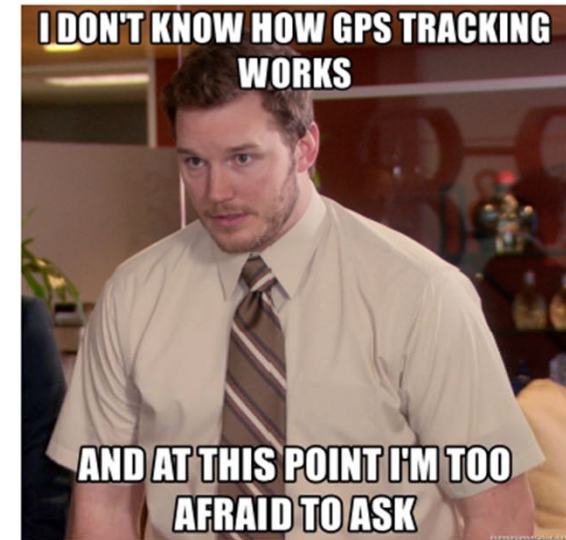
...so this is just a scalar

Really all the
same idea!

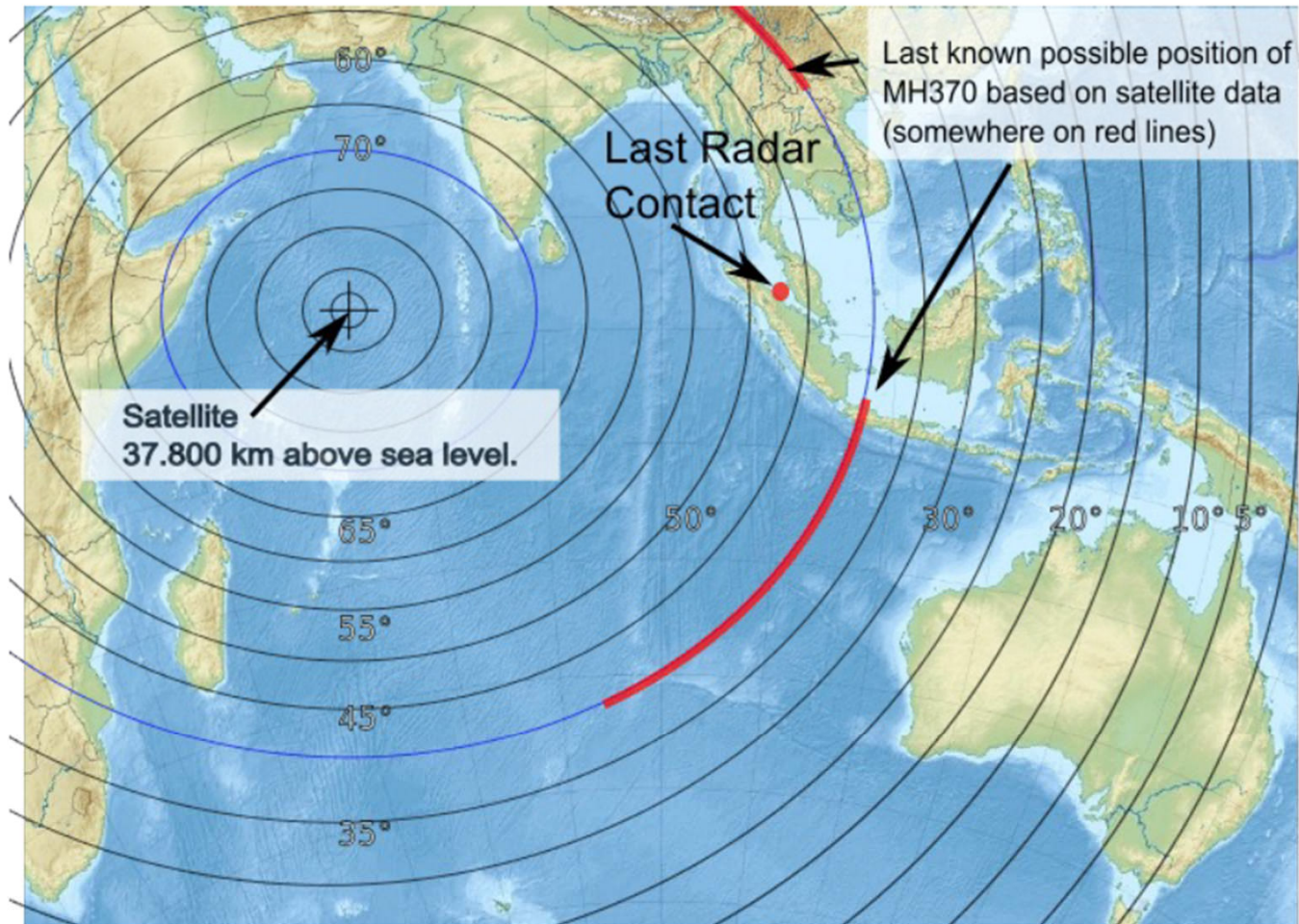
Intense Exercise



Workout time



Tracking in 2-D space



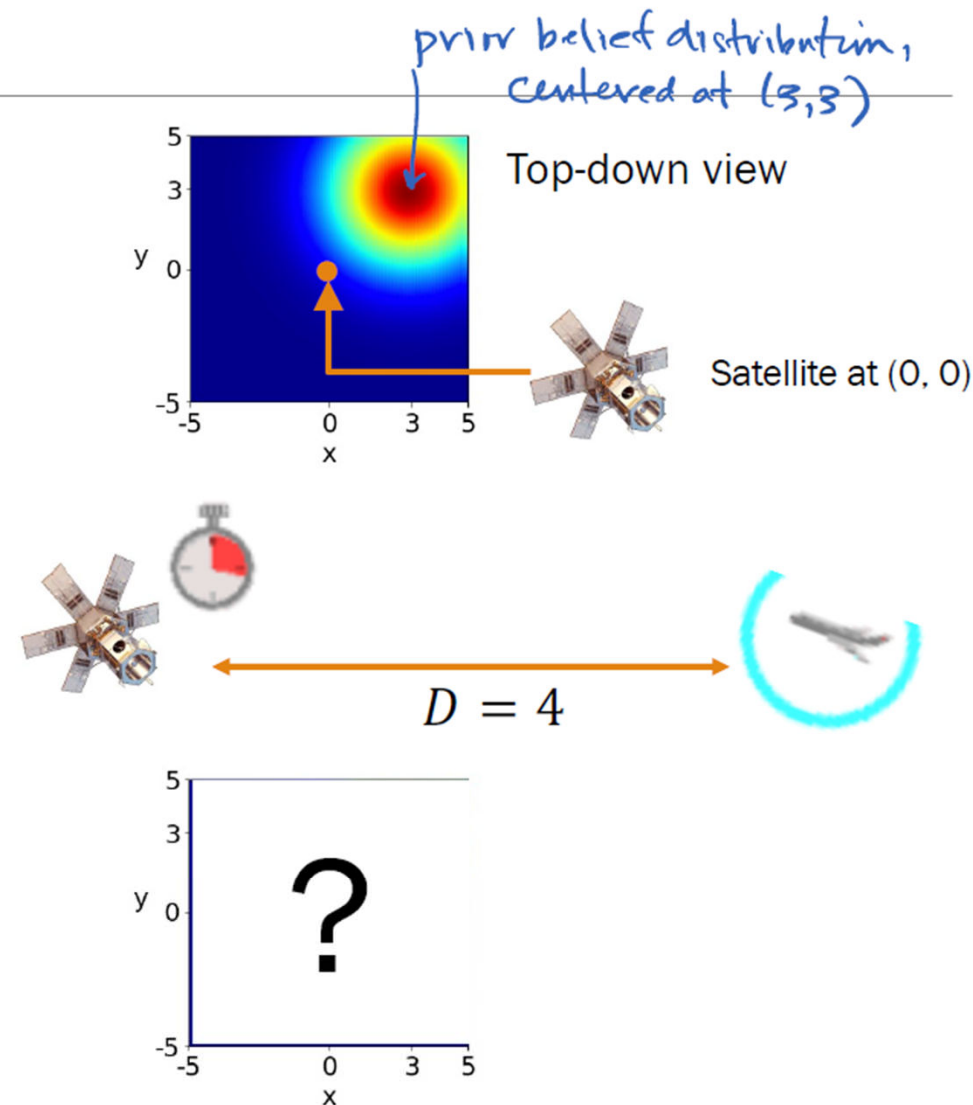
You want to know the 2-D location of an object.

Your satellite ping gives you a noisy 1-D measurement of the distance of the object from the satellite (0,0).

Using the satellite measurement, where is the object?

Tracking in 2-D space

- Before measuring, we have some **prior belief** about the 2-D location of an object, (X, Y) .
- We observe some noisy **measurement** $D = 4$, the Euclidean distance of the object to a satellite.
- After the measurement, what is our **updated (posterior) belief** of the 2-D location of the object?



Tracking in 2-D space

- You have a **prior belief** about the 2-D location of an object, (X, Y) .
- You observe a **noisy distance measurement**, $D = 4$.
- What is your **updated (posterior) belief** of the 2-D location of the object after observing the measurement?

Recall Bayes
terminology:

$$f_{X,Y|D}(x, y|d) = \frac{\overset{\text{likelihood}}{\text{(of evidence)}} \overset{\text{prior}}{\text{belief}} f_{D|X,Y}(d|x, y) f_{X,Y}(x, y)}{\underset{\text{normalization constant}}{f_D(d)}}$$

1. Define prior

$$f_{X,Y|D}(x,y|d) = \frac{f_{D|X,Y}(d|x,y) f_{X,Y}(x,y)}{f_D(d)}$$

You have a **prior belief** about the 2-D location of an object, (X, Y) .

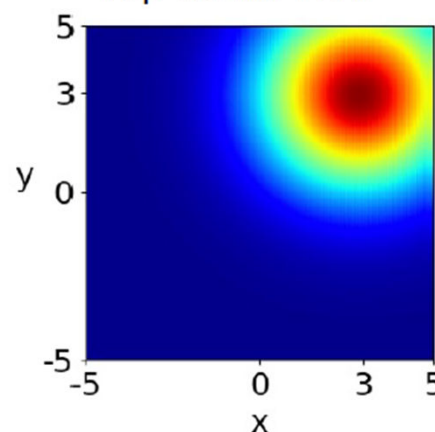
Let (X, Y) = object's 2-D location,
assuming satellite is at $(0,0)$

Suppose the prior distribution is a
symmetric bivariate normal distribution:

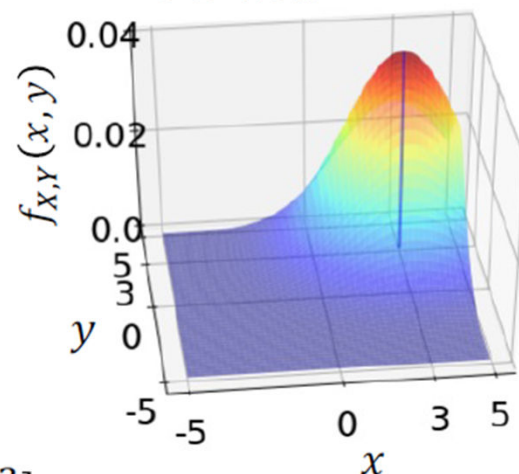
$$f_{X,Y}(x,y) = \frac{1}{2\pi 2^2} e^{-\frac{[(x-3)^2 + (y-3)^2]}{2(2^2)}} = K_1 \cdot e^{-\frac{[(x-3)^2 + (y-3)^2]}{8}}$$

$$\mu = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \Sigma = \begin{bmatrix} 2^2 & 0 \\ 0 & 2^2 \end{bmatrix} \quad \text{normalizing constant}$$

Top-down view



3-D view



2. Define likelihood

$$f_{X,Y|D}(x,y|d) = \frac{f_{D|X,Y}(d|x,y) f_{X,Y}(x,y)}{f_D(d)}$$

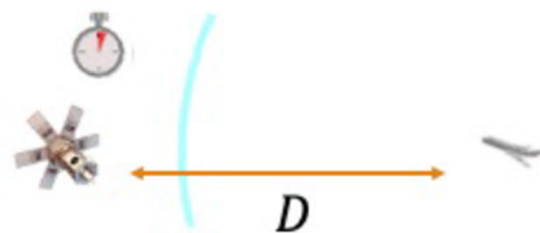
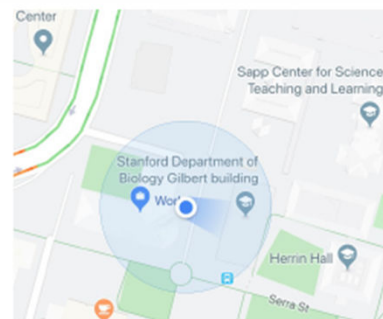
You observe a **noisy distance measurement**, $D = 4$.

If you knew your actual location (x, y) , you could say **how likely** a measurement $D = 4$ is:

Let D = distance from the satellite (radially).

Suppose you knew your actual position: (x, y) .

- D is still noisy! Suppose noise is **standard normal**.
- On average, D is your true Euclidean distance: $\sqrt{x^2 + y^2}$



2. Define likelihood

$$f_{X,Y|D}(x, y|d) = \frac{f_{D|X,Y}(d|x, y) f_{X,Y}(x, y)}{f_D(d)}$$

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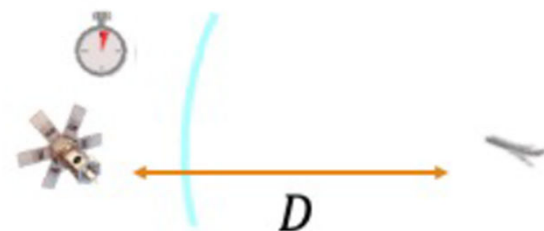
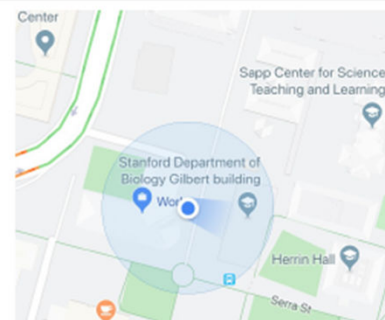
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$$D|X, Y \sim N(\mu = (A), \sigma^2 = (B))$$

$$f_{D|X,Y}(D = d|X = x, Y = y) = \frac{1}{(C) \sqrt{2\pi}} e^{\{(D)\}}$$



2. Define likelihood

$$f_{X,Y|D}(x, y|d) = \frac{f_{D|X,Y}(d|x, y) f_{X,Y}(x, y)}{f_D(d)}$$

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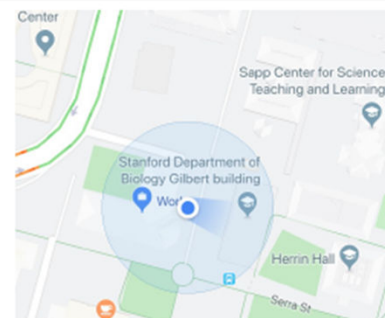
- D is still noisy! Suppose noise is **standard normal**.
- On average, D is your true Euclidean distance: $\sqrt{x^2 + y^2}$

$$D = \sqrt{x^2 + y^2} + Z, Z \sim N(0, 1)$$

$$D|X, Y \sim N\left(\mu = \sqrt{x^2 + y^2}, \sigma^2 = 1\right)$$

$$f_{D|X,Y}(D = d|X = x, Y = y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(d - \sqrt{x^2 + y^2})^2}{2}} = K_2 \cdot e^{-\frac{(d - \sqrt{x^2 + y^2})^2}{2}}$$

normalizing constant



3. Compute posterior

$$f_{X,Y|D}(x, y|d) = \frac{f_{D|X,Y}(d|x, y) f_{X,Y}(x, y)}{f_D(d)}$$

What is your **updated (posterior) belief** of the 2-D location of the object after observing the measurement?

Compute:

Posterior
belief

$$f_{X,Y|D}(x, y|4) = f_{X,Y|D}(X = x, Y = y|D = 4)$$



3. Compute posterior

$$f_{X,Y|D}(x, y|d) = \frac{f_{D|X,Y}(d|x, y) f_{X,Y}(x, y)}{f_D(d)}$$

What is your **updated (posterior) belief** of the 2-D location of the object after observing the measurement?

Compute:

Posterior
belief

$$f_{X,Y|D}(x, y|4) = f_{X,Y|D}(X = x, Y = y|D = 4)$$

Know:

Prior
belief

$$f_{X,Y}(x, y) = K_1 \cdot e^{-\frac{[(x-3)^2 + (y-3)^2]}{8}}$$

Observation
likelihood

$$f_{D|X,Y}(d|x, y) = K_2 \cdot e^{\frac{-(d - \sqrt{x^2 + y^2})^2}{2}}$$

Tips

- Use Bayes' Theorem!
- $f_D(4)$ is just a scaling constant. Why?
- How can we approximate the final scaling constant with a computer?

Tracking in 2-D space

What is your **updated (posterior) belief** of the 2-D location of the object after observing the measurement?

$$\begin{aligned} f_{X,Y|D}(X = x, Y = y | D = 4) &= \frac{\overset{\text{likelihood of } D = 4}{f_{D|X,Y}(D = 4 | X = x, Y = y)} \overset{\text{prior belief}}{f_{X,Y}(x, y)}}{f(D = 4)} \quad \text{Bayes' Theorem} \\ &= \frac{K_2 \cdot e^{-\frac{(4 - \sqrt{x^2 + y^2})^2}{2}} \cdot K_1 \cdot e^{-\frac{[(x-3)^2 + (y-3)^2]}{8}}}{f(D = 4)} \\ &= \frac{K_3 \cdot e^{-\left[\frac{(4 - \sqrt{x^2 + y^2})^2}{2} + \frac{[(x-3)^2 + (y-3)^2]}{8}\right]}}{f(D = 4)} \\ &= K_4 \cdot e^{-\left[\frac{(4 - \sqrt{x^2 + y^2})^2}{2} + \frac{[(x-3)^2 + (y-3)^2]}{8}\right]} \end{aligned}$$

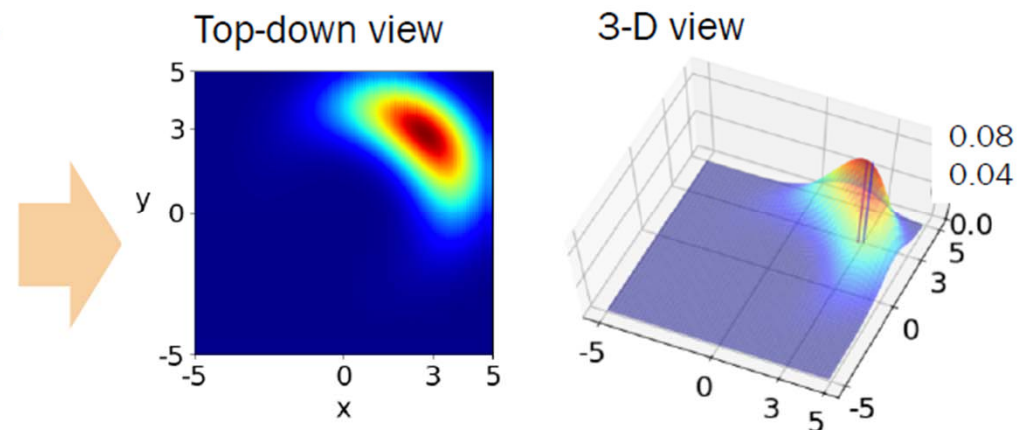
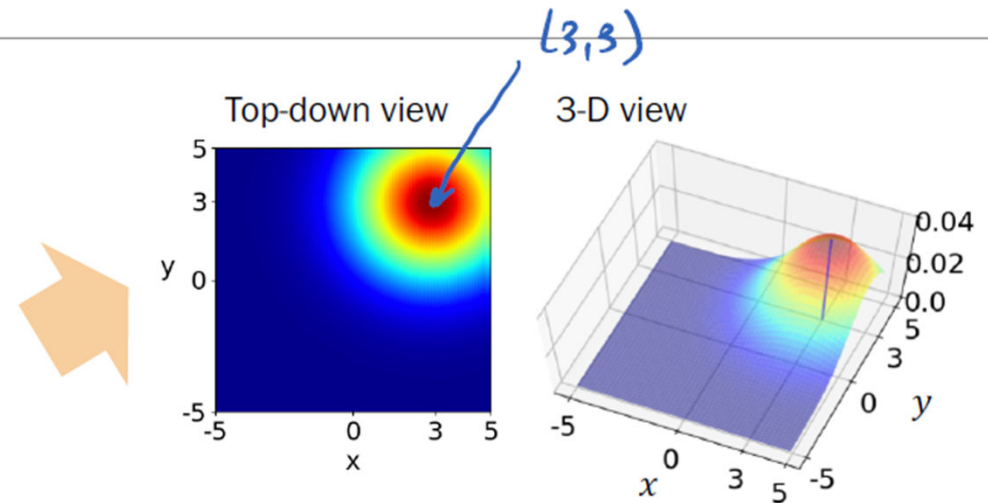
Key: Once we know the part dependent on x, y , we can computationally approximate K_4 so that $f_{X,Y|D}$ is a valid PDF.

Tracking in 2-D space

$(3,3)$ is at $3\sqrt{2} \approx 4.2$ from origin

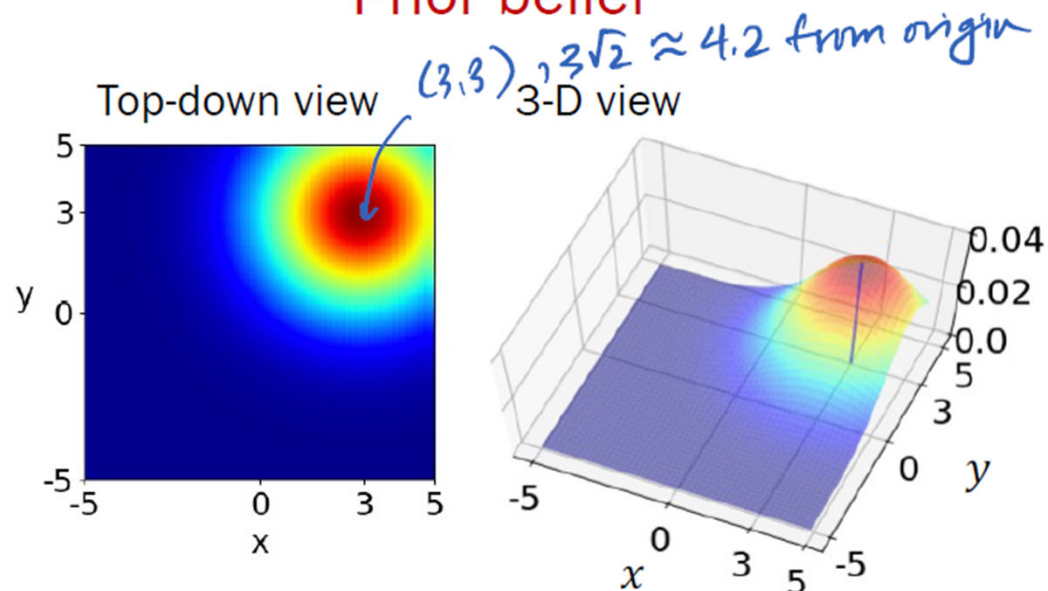
With this continuous version of Bayes' theorem, we can explore new domains.

- Before measuring, we have some **prior belief** about the 2-D location of an object, (X, Y) .
- We observe some noisy **measurement** of the distance of the object to a satellite.
- After the measurement, what is our **updated (posterior) belief** of the 2-D location of the object?



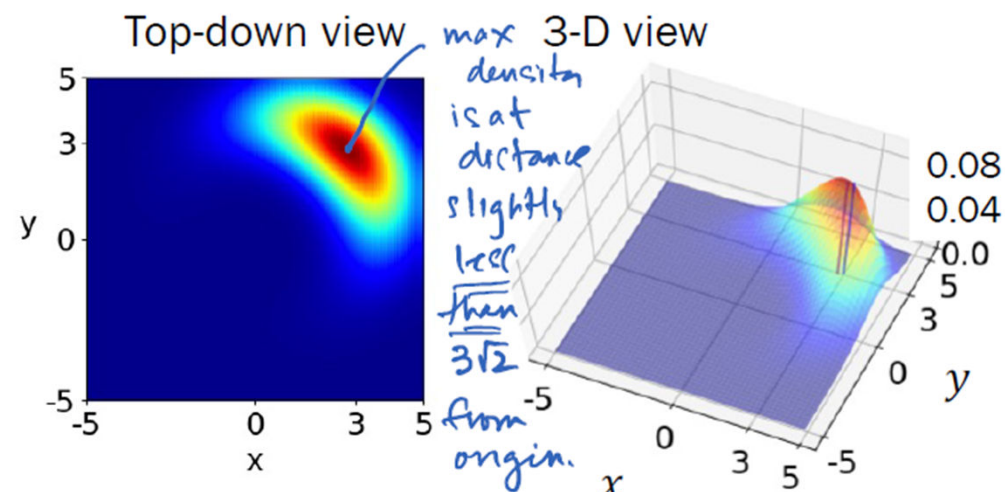
Tracking in 2-D space: Posterior belief

Prior belief



$$f_{X,Y}(x, y) = K_1 \cdot e^{-\frac{[(x-3)^2 + (y-3)^2]}{8}}$$

Posterior belief



observing a 4 indicates prior center at distance of $3\sqrt{2} \approx 4.2$ was too big and should be adjusted

$$f_{X,Y|D}(x, y|4) = K_4 \cdot e^{-\left[\frac{(4 - \sqrt{x^2 + y^2})^2}{2} + \frac{[(x-3)^2 + (y-3)^2]}{8} \right]}$$

How'd you compute that K_4 ?

To be a valid conditional PDF, $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y|D}(x,y|4) dx dy = 1$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K_4 \cdot e^{-\left[\frac{(4-\sqrt{x^2+y^2})^2}{2} + \frac{[(x-3)^2+(y-3)^2]}{8}\right]} dx dy = 1$$

➡ $\frac{1}{K_4} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\left[\frac{(4-\sqrt{x^2+y^2})^2}{2} + \frac{[(x-3)^2+(y-3)^2]}{8}\right]} dx dy$ (pull out K_4 , divide)

Approximate:

$$\frac{1}{K_4} \approx \sum_y \sum_x e^{-\left[\frac{(4-\sqrt{x^2+y^2})^2}{2} + \frac{[(x-3)^2+(y-3)^2]}{8}\right]} \Delta x \Delta y$$

Use a computer!